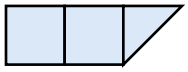
Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q09 (v1)

Filling a shape with pieces: start where there is least room (corner, narrow arm), match slanted edges, check with the number of squares

Warm-up: small steps first (no points)

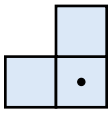
W1. How many half squares (triangles made by cutting a square in two) make up this piece?



A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

W2. Two pieces fill this shape.

Which piece lies on the dot? The pieces may be turned.

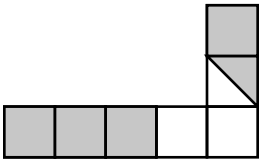


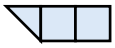
A. ***B.*** ***C.*** ***D.*** ***E.***

Part A: Each question is worth 3 points

1. Some pieces are already in the shape (grey).

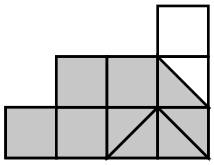
Which piece fills the white gap exactly? The piece may be turned.



A. ***B.*** ***C.***  ***D.*** ***E.***

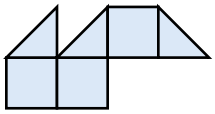
2. Some pieces are already in the shape (grey).

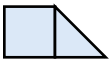
Which piece fills the white gap exactly? The piece may be turned.



A. ***B.*** ***C.*** ***D.*** ***E.***

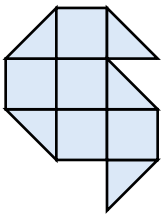
3. How many pieces like the small one are needed to fill the big shape completely? The pieces may be turned.

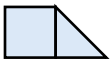




A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 9

4. How many pieces like the small one are needed to fill the big shape completely? The pieces may be turned.



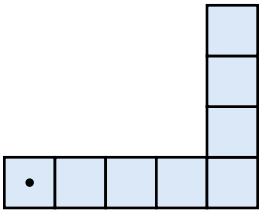


A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 15

Part B: Each question is worth 4 points

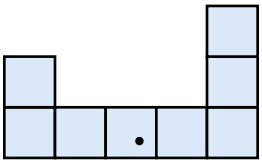
5. Using the pieces A, B, C, D and E you can fill this shape completely. The pieces may be turned.

Which of the pieces lies on the dot?



A. ***B.*** ***C.*** ***D.*** ***E.***

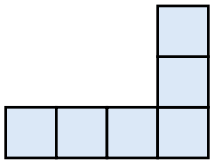
6. Using the pieces A, B, C, D and E you can fill this shape completely. The pieces may be turned. Which of the pieces lies on the dot?



A. ***B.*** ***C.*** ***D.*** ***E.***

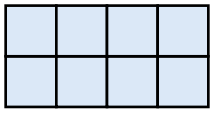
Part C: Each question is worth 5 points

7. Four of the pieces A, B, C, D and E fill this shape completely. The pieces may be turned. Which piece is left over?



A. ***B.*** ***C.*** ***D.*** ***E.***

8. Four of the pieces A, B, C, D and E fill this shape completely. The pieces may be turned. Which piece is left over?



A. ***B.*** ***C.*** ***D.*** ***E.***

Answer Key

W1.	**B**	**W2.**	**B**	**1.**	**D**	**2.**	**A**	**3.**	**C**	**4.**	**C**
5.	**C**	**6.**	**C**	**7.**	**B**	**8.**	**C**				

Solutions

W1. B (5) — A whole square is 2 half squares. Counting gives 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · w_count_halves]

W2. B — Start in the corner with the dot and try each piece. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · w_corner_first]

1. D — Count the half squares of the gap (5) and look at where its slanted edges are. Only piece D has the same shape. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · missing_piece]

2. A — Count the half squares of the gap (3) and look at where its slanted edges are. Only piece A has the same shape. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · missing_piece]

3. C (3) — Count half squares: the shape has 9, one piece has 3. 9 ÷ 3 = 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · how_many_tiles]

4. C (5) — Count half squares: the shape has 15, one piece has 3. 15 ÷ 3 = 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · how_many_tiles]

5. C — Start where there is the least room: the corners and the narrow parts of the shape. Slanted edges must meet slanted edges. When all pieces are in, piece C lies on the dot. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · dot_piece]

6. C — Start where there is the least room: the corners and the narrow parts of the shape. Slanted edges must meet slanted edges. When all pieces are in, piece C lies on the dot. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · dot_piece]

7. B — Pieces A and B have the same number of half squares, so counting alone does not decide. Fit the pieces in, starting in the corners: piece B is not needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · left_over]

8. C — Pieces E and C have the same number of half squares, so counting alone does not decide. Fit the pieces in, starting in the corners: piece C is not needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q9 · left_over]